- **Exogenous variables** $\mathbf{U}$ are inputs sampled from $P_{\mathbf{U}}$, representing the behavior we seek to explain.
- **Structural assignments** $\mathbf{f}$ are the deterministic functions defining the network's computation (layers, attention mechanisms, nonlinearities).
- **Causal graph** $\mathcal{G}$: the network's computation graph.

The SCM induces a unique *observational distribution* over $\mathbf{V}$: sampling corresponds to drawing inputs from $P_{\mathbf{U}}$, executing a forward pass, and recording desired activations. Also, the SCM encodes *interventional* and *counterfactual* distributions based on external modifications of the inner computation. This perspective is standard within mechanistic interpretability (Olah et al., 2018; Cammarata et al., 2020; Geiger et al., 2022b; 2025) and is illustrated in Figure 2. Then, a *causal query* is any well-specified quantity about $\mathfrak{C} := (f, P_{\mathbf{U}})$, such as "What distribution would the network produce if we forced activation $V_i$ to value $v$?" or "How much does attention head $V_a$ causally contribute to correct factual recall?"

### 4.3 Explanations as Surrogate Models

In an attempt to provide a general statistical-causal perspective on interpretability, we formalize explanations as *surrogate models*: simpler computational descriptions designed to answer chosen collections of causal queries about a target system. This perspective treats interpretability as a form of model compression, where we seek a simpler model that faithfully approximates a complex system's behavior for queries we care about. In this perspective, every interpretability method is characterized by three ingredients:

1. **Query space** $Q$ with **distribution** $\mu$: The set of causal queries to be answered by the explanation. It dictates what aspects of $\mathfrak{C}$ should be explained. This encodes our explanatory goals.
2. **Surrogate class** $\mathcal{E}$: The class of admissible explanations. It dictates what forms the explanation can take, e.g., circuits, sparse subgraphs, linear probes, concept vectors, causal graphs, ...
3. **Discrepancy measure** $D$: How we measure whether a surrogate (member of $\mathcal{E}$) *correctly* answers queries.

This framework is (non-rigorously) illustrated with the example of circuit discovery in Figure 3. While standard causal inference often concentrates $\mu$ on a *single* causal query (e.g., average treatment effect), interpretability aims to answer *many diverse queries* drawn from a non-trivial distribution $\mu$. For example, $\mu$ might distribute probability over interventional queries and counterfactual queries across network components, or any functionals of the interventional and counterfactual distributions.

For instance, we can view each candidate explanation $e \in \mathcal{E}$ as defining a *query-answering map* $S_e : Q \to \mathcal{R}$, where $S_e(q)$ is the surrogate's predicted answer to query $q$, and $\mathcal{R}$ is the space of possible answers for query $q$ (e.g., probability distributions, scalar effects, or discrete predictions). The surrogate's *fidelity* is measured by the discrepancy function $D : \mathcal{R} \times \mathcal{R} \to \mathbb{R}_+$ quantifying the error between the answer from $q(\mathfrak{C})$ and the surrogate's prediction $S_e(q)$. Then, we can define the *population risk* of a candidate explanation as the expected error over queries:

$$L_\mu(e) = \mathbb{E}_{q\sim\mu}\big[D\big(q(\mathfrak{C}),\, S_e(q)\big)\big]. \tag{2}$$

An ideal explanation $e^* \in \mathcal{E}$ minimizes this risk: $e^* \in \arg\min_{e\in\mathcal{E}} L_\mu(e)$.

**Interpretability Task and Identifiability.** We call the triple $(\mu, \mathcal{E}, D)$ an *interpretability task*, fully specifying what we aim to explain (query distribution $\mu$), what explanations are admissible (hypothesis class $\mathcal{E}$), and how we measure success (discrepancy $D$). The task is *identifiable* if $L_\mu$ admits a unique minimizer in $\mathcal{E}$ (potentially up to predefined acceptable symmetries). Identifiability captures whether the surrogate class can, in principle, be distinguished using the queries deemed relevant by $\mu$. Without identifiability, multiple incompatible explanations achieve the same population risk, making inference fundamentally ambiguous. The analysis of Section 2 indicates that the most common tasks are not identifiable. Finally, this reframing highlights that explanations are

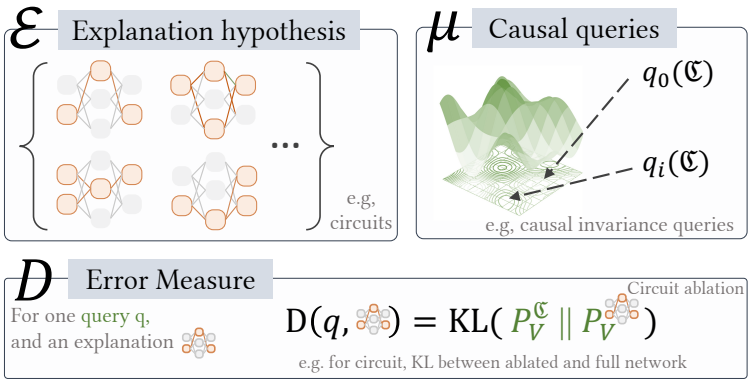



Figure 3: An interpretability task is defined by three elements: $\mathcal{E}$, the hypothesis space; $\mu$, the distribution over causal queries about the SCM (model and behavior); and $D$, the error measure.

*pragmatic computational summaries* of the structure encoded by $\mathfrak{C} := (f, P_{\mathbf{U}})$. They are inferred models useful for specific explanatory purposes.

**Estimation with Finite Data.** In practice, we face two types of finite sampling difficulties. First, we observe only finitely many queries $q_1, \ldots, q_n \sim \mu$ from the query distribution. Second, for each query $q_j$, we can only collect a finite amount of computational traces by sampling inputs $\mathbf{U} \sim P_{\mathbf{U}}$ and recording the corresponding activations and outputs, potentially under interventions. Let $\mathcal{T}_n$ denote the complete dataset of query-trace pairs. An interpretability method $M$ acts as an *estimator*, mapping this finite dataset to an estimated surrogate explanation:

$$\hat{e} := M(\mathcal{T}_n). \tag{3}$$

A natural estimator is given by empirical risk minimization: choose $\hat{e}$ to minimize the empirical risk $\widehat{L_\mu}(e) = \frac{1}{n}\sum_{j=1}^{n} \widehat{D}(q_j(\mathfrak{C}),\, S_e(q_j))$, where $\widehat{D}$ is estimated from finite traces.

Relevant to this exposition, Senetaire et al. (2023) proposed a statistical framing for feature attribution. Then, previous works already explored hypothesis testing and uncertainty quantification for circuit discovery (Shi et al., 2024; Méloux et al., 2025).

### 4.4 Re-Interpreting Documented Issues

The framework does not prescribe which $(\mu, \mathcal{E}, D)$ researchers should adopt. Rather, it provides a *shared language* for making assumptions explicit and rooted in the tools of statistical inference. Different research programs will choose different hypothesis classes or query distributions; the framework ensures that such choices are transparent and their implications are analyzable. Table 1 in the appendix illustrates how existing interpretability methods can be mapped into this formulation, each implicitly making assumptions about queries, surrogates, and error metrics. Under this view, the issues described in Section 2 can be understood as problems of *non-identifiability*.

**Behavioral Benchmarks.** Benchmarks that evaluate model outputs against a gold standard (averaged success over input distributions) ask an identifiable question: *how well does the model perform under a specific task distribution and error metric?* This is arguably the simplest form of interpretability and drove most of the progress in AI. Its usefulness depends on the construction of the benchmark, but the inference problem is well-posed.

**Concept-Based Approaches.** Predictive methods (probes, SAEs) inherit non-identifiability issues from the underlying underspecification of machine learning tasks (D'Amour et al., 2022). For example, methods like Concept Activation Vectors (Kim et al., 2018; Cunningham et al., 2023) postulate that internal states $\mathbf{v}$ are generated by interpretable concepts $\mathbf{z}$ via $\mathbf{v} = g(\mathbf{z})$. This is an instance of (causal) representation learning, which is **non-identifiable** without auxiliary information (Locatello et al., 2019; Khemakhem et al., 2020). It is therefore unsurprising that proposed improvements

mirror standard remedies for underspecification in machine learning: regularization in the form of capacity control for probes (Belinkov, 2022) or cross-validation to assess generalization (Kantamneni et al., 2025).

**Causal Mediation Analysis.** Methods like causal mediation analysis (Meng et al., 2022; Vig et al., 2020a) estimate the indirect effect of a component on observed outputs. As the intervention and model are fully specified, the mediation estimand is unique and **identifiable**. However, the explanatory claim that a component with high effect is the *locus* of a mechanism is **not identifiable**, because of the overdetermined causal structure.

**Circuit Discovery and Causal Approaches.** Circuit discovery seeks a subgraph $G' \subset G$ that preserves the model's performance. This task faces the "Hydra effect" (McGrath et al., 2023) and causal overdetermination. If parallel pathways $A$ and $B$ are sufficient, circuits containing only $A$ or only $B$ both satisfy fidelity criteria. Thus, even *correct* causal methods may recover many different explanations consistent with the same behavior (Méloux et al., 2025). Addressing this requires formulating identifiable causal questions. Causal abstraction (Beckers & Halpern, 2019; Geiger et al., 2025) offers a promising direction, as it operates at a coarser representational level where overdetermination can be absorbed into the abstracted representations. However, current operational metrics demonstrate empirical non-identifiability (Méloux et al., 2025; Sutter et al., 2025).

## 5 Discussion

The systematic failures documented in Section 2 demanded an explanation. We have argued that these pathologies share a common root cause: **non-identifiability**. Most current interpretability tasks attempt to infer explanatory structures that are not uniquely determined by available computational traces. To trace a path forward, we proposed one formalization of interpretability as statistical-causal inference. This framework is tentative rather than definitive; we encourage the community to improve upon it. The important aspect is the *methodological commitment* to making assumptions explicit and quantifying uncertainty rigorously.

### 5.1 Advantages of the Statistical-Causal Perspective

Drawing on the philosophy of science (Chang, 2004; Woodward, 2003; Potochnik, 2017) and recent calls for a pragmatic approach to interpretability (Davies & Khakzar, 2024; Williams et al., 2025), the framework naturally distinguishes the *explanandum* (what is to be explained, encoded in $\mu$) from the *explanans* (what does the explaining, encoded in $\mathcal{E}$). Researchers and practitioners have substantial freedom in choosing both. There is no single "correct" explanation of a neural network. The appropriate type of description depends on one's purposes (Potochnik, 2017). Descriptive understanding corresponds to queries about observational distributions; predictive goals involve queries requiring surrogates to generalize to new input distributions; control and intervention require queries about counterfactual or interventional distributions.

However, once the explanatory project is specified, i.e., once $(\mu, \mathcal{E}, D)$ are fixed, the explanation becomes an **objective inference problem**. The best surrogate $e^* \in \mathcal{E}$ is the one minimizing $L_\mu(e)$, and is a property of the system itself and the interpretability task. If the task is identifiable, this explanation is unique (up to permissible symmetries, e.g., rotation invariance in representation space). This reconciles pluralism about explanatory goals with rigor about explanatory claims.

Perhaps most critically, the statistical framing demands that interpretability methods report not just point estimates but *confidence sets* or *posterior distributions* over explanations (in case of Bayesian framing). Just as we would not trust a clinical trial reporting effect sizes without confidence intervals, we may not trust interpretability claims without uncertainty quantification. When explanations are non-identifiable, this uncertainty will be large; when they are identifiable with finite data, uncertainty shrinks as observations accumulate.